The empirical recurrence for OEIS A224387: recovering a conjecture that is not written down

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Abstract

OEIS A224387 records "Empirical recurrence of order 47", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 245$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 47, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 47 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A224387 is "Number of nX4 0..3 arrays with diagonals and antidiagonals unimodal and rows nondecreasing.". It has offset 1 and begins

$$35, 1225, 27790, 497958, 8573507, 152271025, 2780848289, 51325449985, \ \ldots$$

The entry states:

> Empirical recurrence of order 47 (see link above)

As of the "Last modified" line on the live entry (Oct 03 2025 23:25:36, revision 10) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 8960$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 245$ of the 8960, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 245.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 490$ exact terms of the model returns order 47 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{47} c_i\, a(n-i),$$

c_1	35	c_{13}	−9353465802	c_{25}	6056096808956027	c_{37}	29193203943363430620
c_2	−347	c_{14}	22907942961	c_{26}	120529751422045120	c_{38}	−57524461769120064408
c_3	201	c_{15}	−518156900620	c_{27}	296450768341487431	c_{39}	−131928572879374101552
c_4	15835	c_{16}	5443559915134	c_{28}	302823956100189422	c_{40}	−137526064967129366808
c_5	−112969	c_{17}	−12579348188621	c_{29}	−515461505774444462	c_{41}	−88010188468129307376
c_6	345007	c_{18}	32496197107373	c_{30}	−2709615901105085895	c_{42}	−35071926611963610240
c_7	−1505879	c_{19}	−63358298269448	c_{31}	−7165123256767822059	c_{43}	−8803204118572002048
c_8	12343544	c_{20}	162634784498625	c_{32}	−9019593973084585116	c_{44}	−1396776267778002048
c_9	60088575	c_{21}	103727369588831	c_{33}	2206070004168658884	c_{45}	−124536128909241600
c_{10}	−799543580	c_{22}	−3368250523801500	c_{34}	34921228515993195408	c_{46}	−37561989210315264
c_{11}	2868734783	c_{23}	−179504481028262	c_{35}	72879212115105618024	c_{47}	−191146522097664
c_{12}	−8185999780	c_{24}	−9376674067470877	c_{36}	75582607071014516484		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 47, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $47+4$ terms removed returns order 47 as well, so $\deg q_{\min} = 47 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $47+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 47 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A224387.

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[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.